

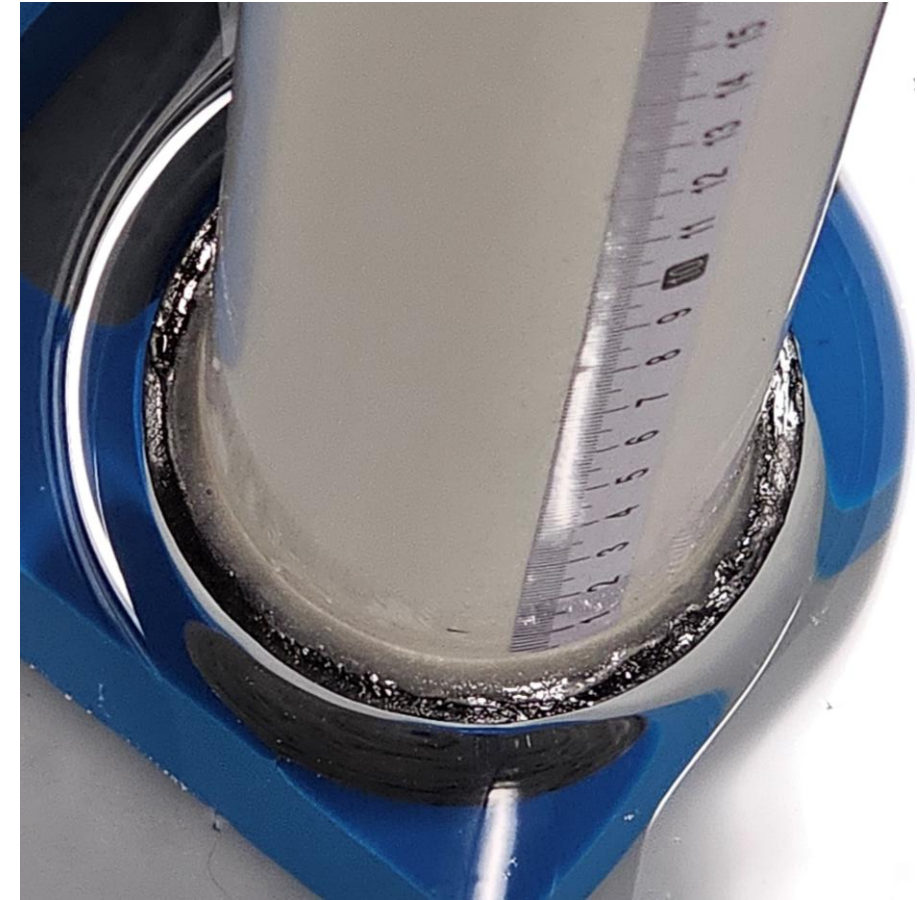


Project Update:
H2 Float Water Level Monitoring System

Kathryn Dunstan

Progress Since Last Meeting

- Calibrated Float Sensor
- Implemented LoRa
- Saving data to SD card
- Battery powered data transmissions

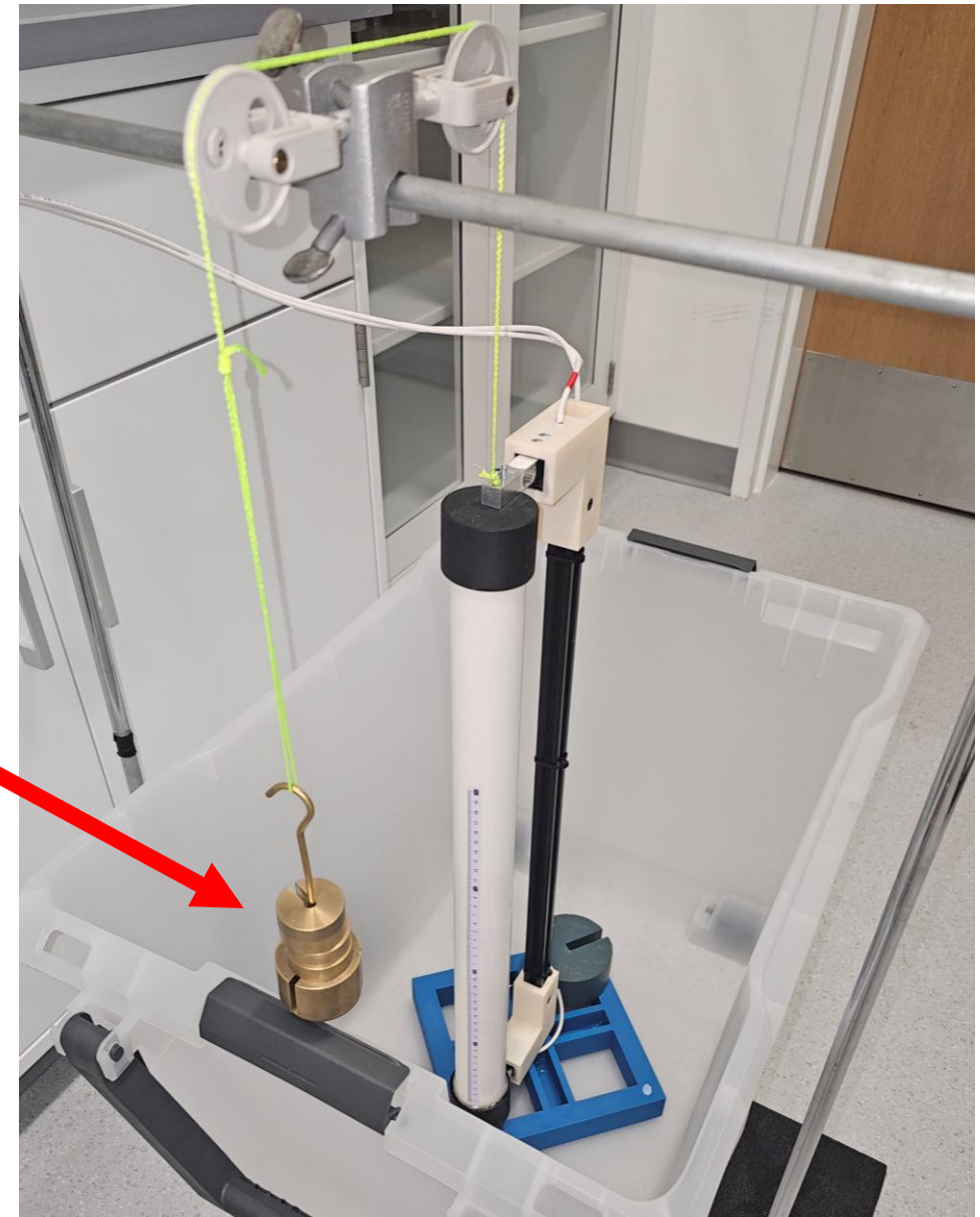


Float Sensor Calibration

Calibration uses...

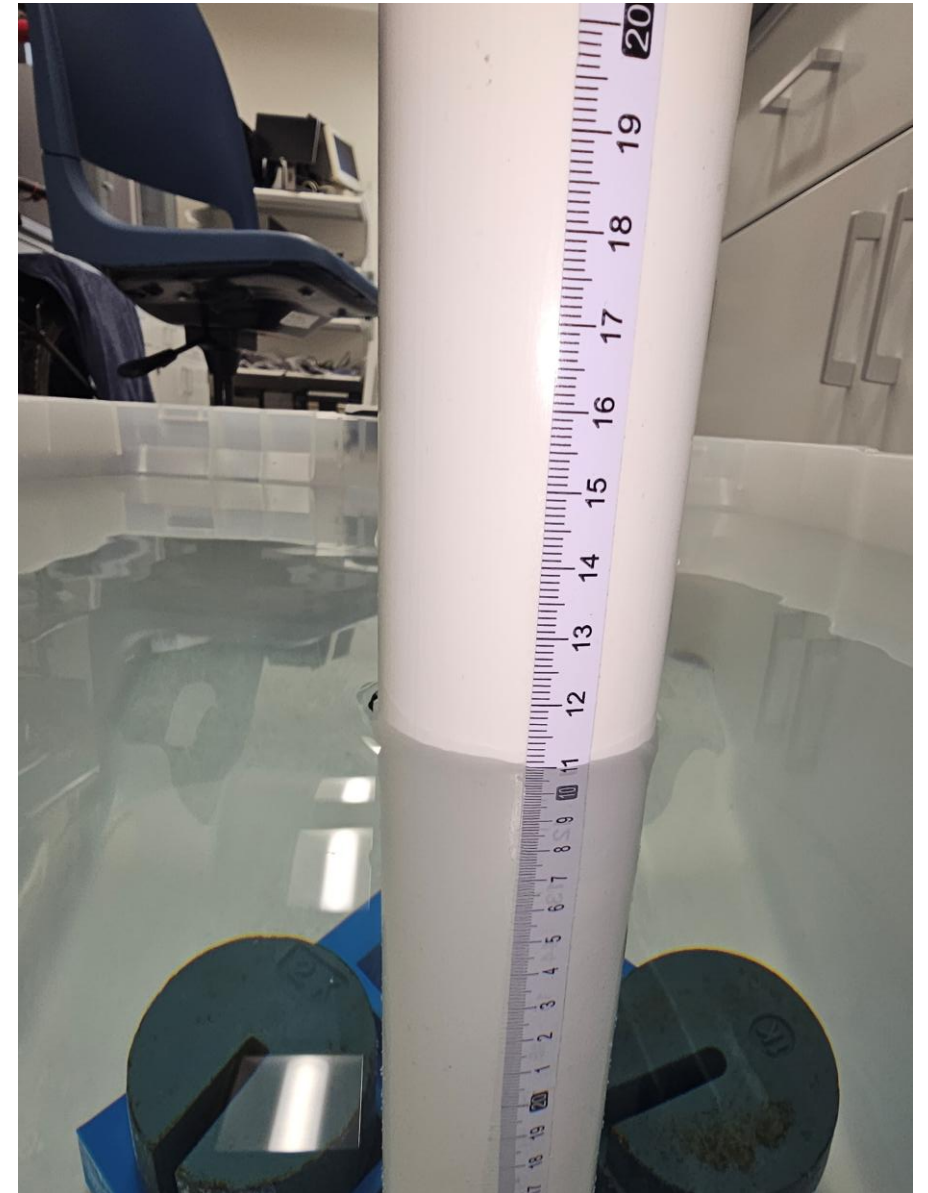
- 25 Samples tare
- 25 Samples 750g (7.4 N)

Upper Tare	Lower Tare	Upper 750g	Lower 750g
Averages			
8598996	8425027	8805932	8060228
Standard Deviation			
5037	5725	2512	3452
%Difference			
0.06%	0.07%	0.03%	0.04%



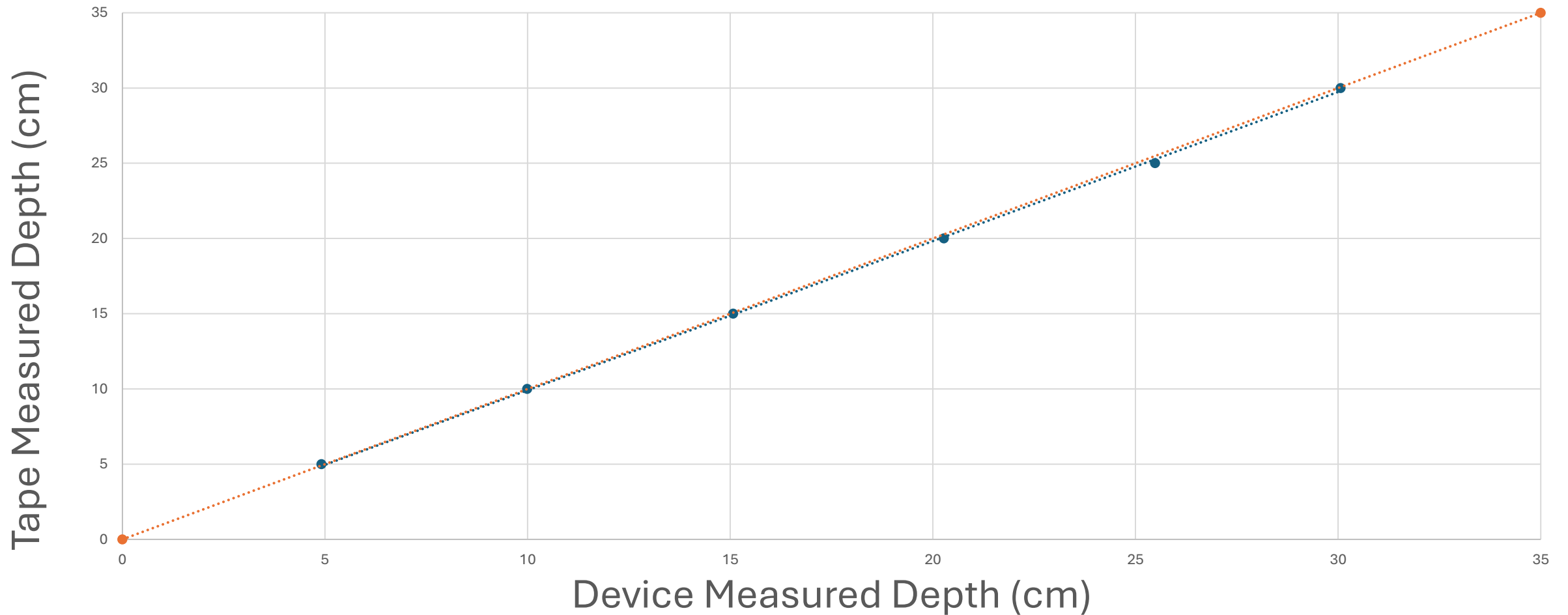
H2 Float Results

Device Measured Depth (cm)	Tape Measured Depth (cm)
4.9	5
10.0	10
15.1	15
20.3	20
25.5	25
30.1	30



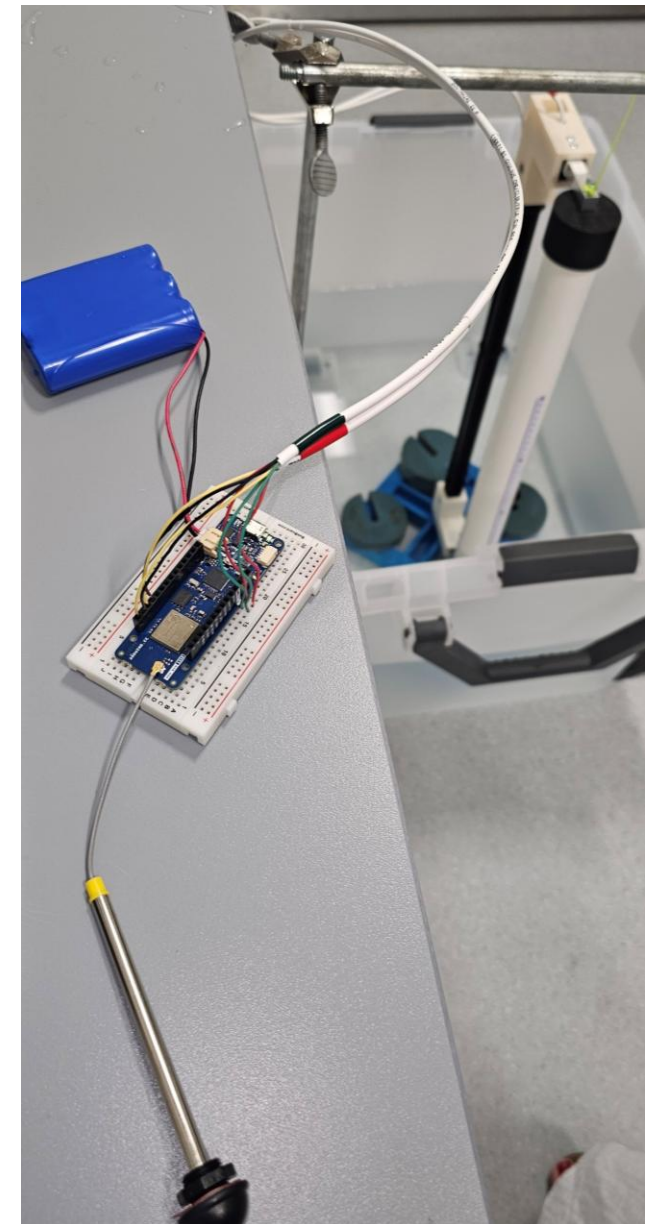
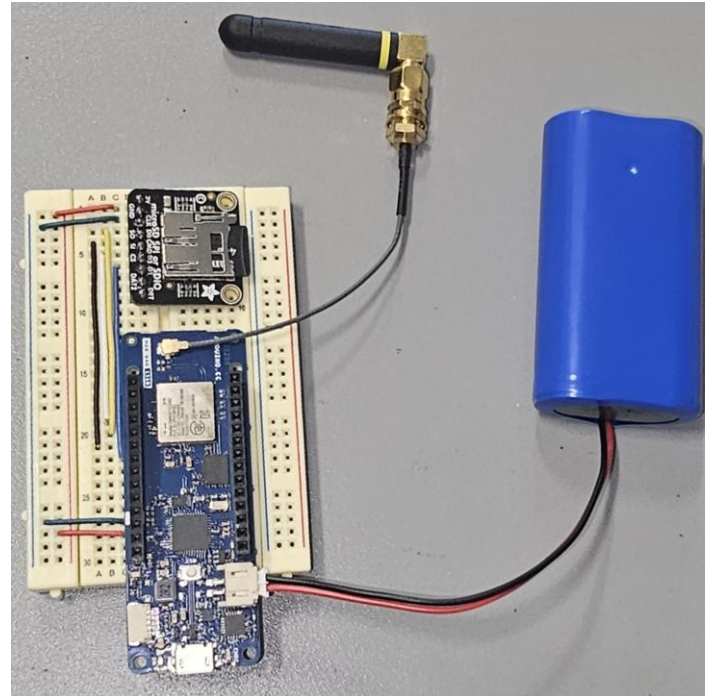
H2 Float Results

Tape Measured Depth vs Device Measured Depth



LoRa Sending and Receiving

- Data Transmission
Sending and receiving
- Battery powered runs
successful
- Data saved to SD Card



Future Plans

- Moving into low power usage
 - LiPo battery
 - Solar
- Time stamping data
- Flume test
- Long term test
- Field test
- Electronics enclosure IP67